Unicode at gigabytes per second

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From characters to bits

Morse code

- A : 0 1
- B : 1 0 0 0
- C : 1 0 1 0

26 letters.

Fixed-length codes

- Baudot code (~1860). 5 bits.
- Hollerith code (~1896). 6 bits.
- American Standard-Code for Information Interchange or ASCII (~1961). 7 bits. 128 characters.

Dec	Hex	Chr	Dec	Hex	Chr	Dec	Hex	Chr	Dec	Hex	Chr
0	00	NUL	32	20	Space	64	40	@	96	60	`
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	|
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	DEL

Too many fixed-length codes!

- IBM: Binary Coded Decimal Interchange Code. 6 bits.
- IBM: Extended Binary Coded Decimal Interchange Code or EBCDIC. 8 bits.
- ISO 8859 (~1987). 8 bits. European.
- Thai (TIS 620), Indian languages (ISCII), Vietnamese (VISCII) and Japanese (JIS X 0201).
- Windows character sets, Mac character sets.

Unicode (late 1980s)

- Extends ASCII.
- Universal.
- Replaces all other standards.
- Typography, full localisation, extensible.

Unicode: how many bits?

- 16 bits ought to be enough?
- Numerical range from 0x000000 to 0x10FFFF.
- Would need 20 to 21 bits.

UTF-16 and UTF-8

Two main formats.

UTF-16: Java, C#, Windows

UTF-8: XML, JSON, HTML, Go, Rust, Swift

UTF-16 and UTF-8

character range	UTF-8 bytes	UTF-16 bytes
ASCII (0000-007F)	1	2
latin (0080-07FF)	2	2
asiatic (0800-D7FF, E000-FFFF)	3	2
supplemental (010000-10FFFF)	4	4

UTF-16

- 16-bit words.
- characters in 0000-D7FF and E000-FFFF, stored as 16-bit values---using two bytes.
- characters in 010000-10FFFF are stored using a 'surrogate pair'.
- Comes in two flavours (little and big endian at the 16-bit level).

UTF-16 (surrogate pair)

- first word in D800-DBFF.
- second word in DC00-DFFF.
- character value is 10 least significant bits of each---second element is least significant.
- add 0x10000 to the result.

UTF-8

- 8-bit words (no endianess)
- One 'leading' byte followed by 0 to 3 bytes.

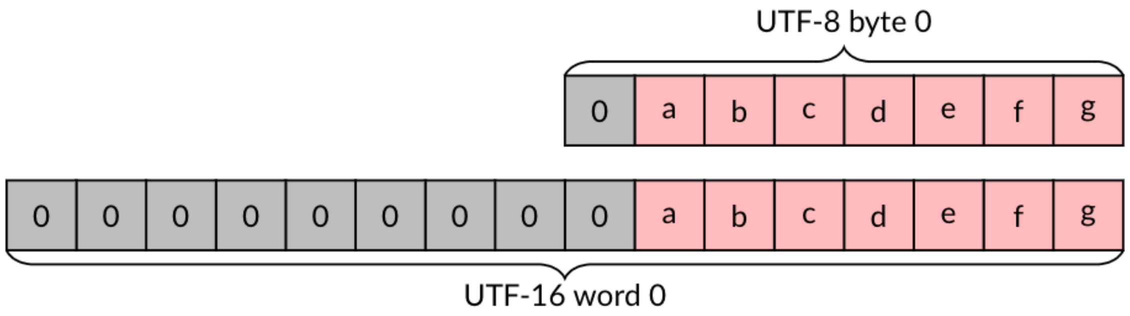
UTF-8 format

- Most significant bit of leading is zero, ASCII: [01000001].
- 3 most significant bits 110, two-byte sequence: [11000100] [10000101].
- 4 most significant bits 1100, three-byte sequence.
- 5 most significant bits 11000, four-byte sequence.
- Non-leading bytes have 10 as the two most significant bits.

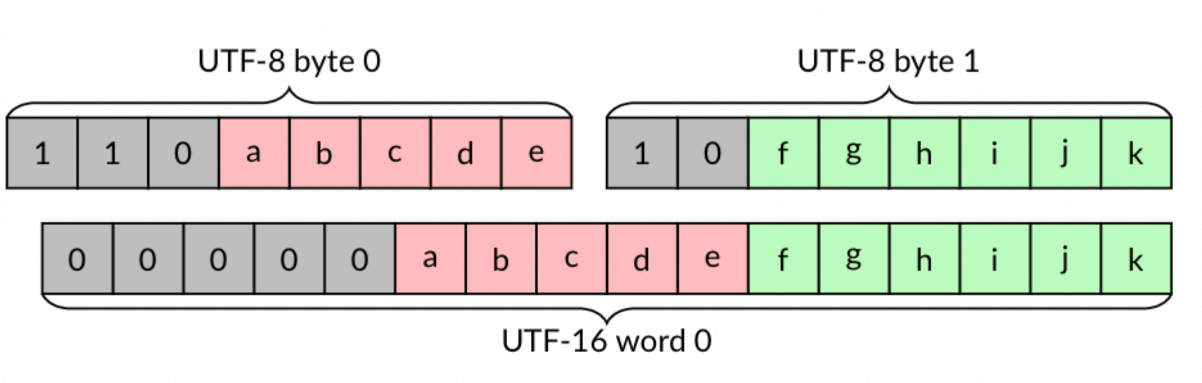
UTF-8 validation rules

- The five most significant bits of any byte cannot be all ones.
- The leading byte must be followed by the right number of continuation bytes.
- A continuation byte must be preceded by a leading byte.
- The decoded character must be larger than 7F for two-byte sequences, larger than 7FF for three-byte sequences, and larger than FFFF for four-byte sequences.
- The decoded code-point value must be less than 110000
- The code-point value must not be in the range D800-DFFF.

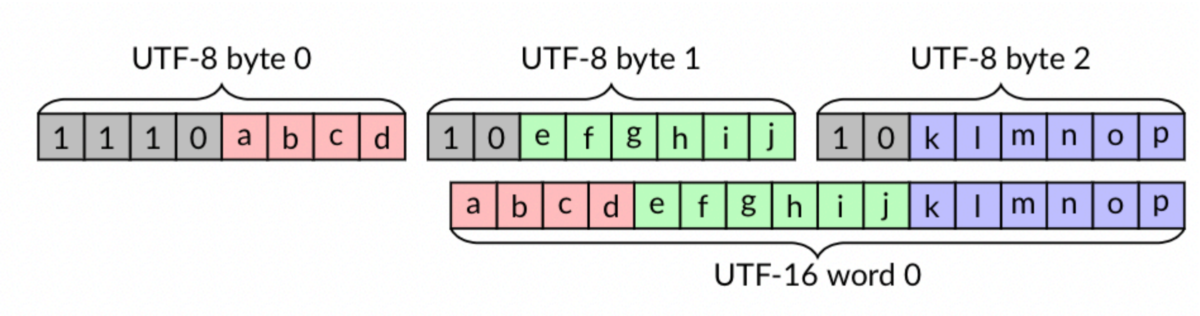
UTF-8/UTF-16 comparison (ASCII)



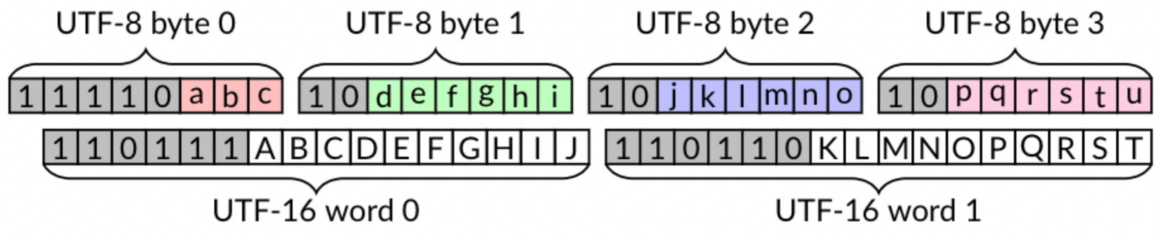
UTF-8/UTF-16 comparison (2-bytes)



UTF-8/UTF-16 comparison (3-bytes)



UTF-8/UTF-16 comparison (4-bytes)



UTF-8/UTF-16 transcoding

- Must convert (transcode) from one format to the other format, while validating the input.

Some numbers

- bandwidth between node instances: over 3 GB/s
- PCIe 4.0 disks (and PlayStation 5): over 5 GB/s
- Popular C++ trancoding library (ICU): ~1 GB/s

Gigabytes per second?

- x64, ARM, POWER: have SIMD instructions.

	UTF-8 to UTF-16	UTF-16 to UTF-8	validation	table size
Cameron's u8u16 (2008)	yes	no	yes	N/A
Inoue et al. (2008)	partial	no	no	105 kB
simdutf	yes	yes	yes	20 kB

Software implementations (no formal paper): Goffart (2012) and Gatilov (2019)

Vectorized permutation

- Can permute blocks of 16 bytes (or 32 bytes) using a single cheap instruction.
- Need a precomputed shuffle mask.
- data : [a b c d e f g]
- shuffle mask : [3 1 0 3 3 2 -1] (indexes)
- result : [d b a d d c 0]
- Conversely may be used as a form of vectorized table lookup.

UTF-8 to UTF-16 transcoding (core)

- Take a block of bytes.
- Continuation bytes (leading bits 10, less than -64)
- Non-continuation bytes are leading bytes
- Bytes before a leading byte end a character
- Build a bitmap
- Use the bitmap in a lookup table

UTF-8 to UTF-16 transcoding (example)

Start with...

[01000001] ([11000100] [10000101])
[01100011] ([11000011] [10000011]) [01101100] ([11000101] [10111010])

We have 9 bytes. Build a 9-bit bitmap where '1' means the end of a character

101101101

Use this as index in a table.

UTF-8 to UTF-16 transcoding (table)

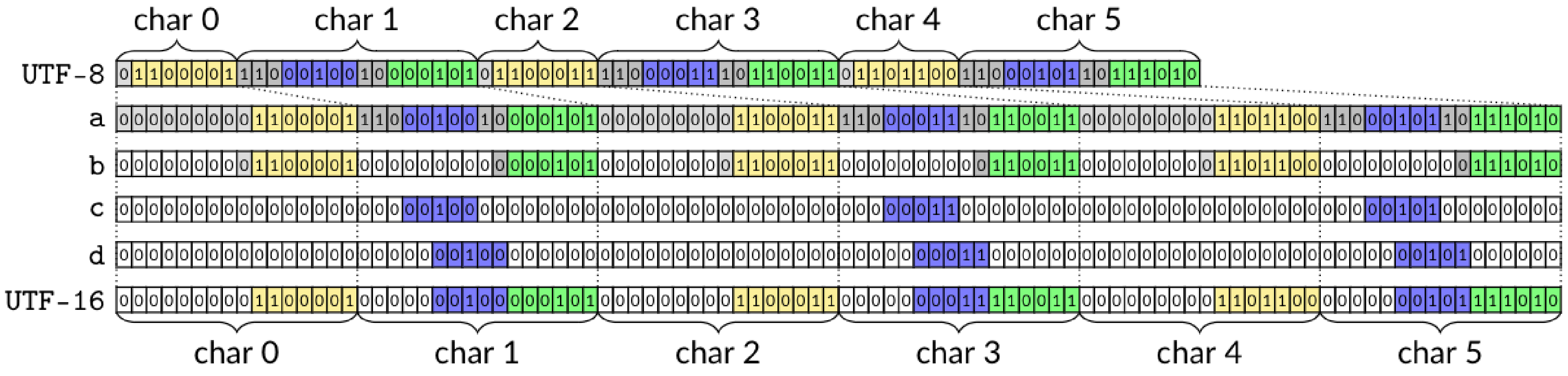
- If using 12-byte blocks, need 4096-long table.
- Each entry points to a shuffle mask and number of consumed bytes.

UTF-8 to UTF-16 transcoding (cases)

Shuffle masks are sorted into 'cases'.

1. First 64 cases correspond to 1-byte or 2-byte characters only.
2. Next 81 cases correspond to 1, 2 or 3 bytes per character.
3. Next 64 cases correspond to general case (1 to 4 bytes).

Each case corresponds to a code path.


char 0
char 1
char 2
char 3
char 4
char 5
UTF-8
a
b
c
d
UTF-16
char 0
char 1
char 2
char 3
char 4
char 5

UTF-8 to UTF-16 transcoding (more tricks)

1. Load blocks of 64 bytes.
2. Check for fast paths (e.g. all ASCII).
3. Eat 12 bytes at a time within 64 bytes.
4. Add a few fast path (e.g., all ASCII, all 2-byte, all 3-byte).

UTF-8 to UTF-16 transcoding (validation)

Given a 64-byte block, we can use a fast vectorized validation routine.

- Validating UTF-8 In Less Than One Instruction Per Byte, Software: Practice and Experience 51 (5), 2021

UTF-8 to UTF-16 transcoding (core algo)

- You can identify most UTF-8 errors by looking at sequences of 3 nibbles (4-bit).
- E.g., ASCII followed by continuation, leading not followed by continuation byte.

Do three lookups (using shuffe mask) and compute a bitwise AND. We call this vectorized classification.

Simplified vectorized classification

- Suppose you want to find all instances where value 3 is followed by value 1 or 2.
- Create two lookup tables.
- One for first nibble [0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0]
- second nibble [0,1,1,0,0,0,0,0,0,0,0,0,0,0,0,0]
- Lookup first nibble in table, lookup second, compute bitwise AND.
- If result is 1, you have a match.
- Can do this in parallel over many values.

Fancier vectorized classification

- Suppose you want to find all instances where value 3 is followed by value 1 or 2. Value 5 followed by 0. Value 6 followed by 10.
- Create two lookup table2.
- One for first nibble [0,0,0,1,0,2,4,0,0,0,0,0,0,0,0,0]
- second nibble [2,1,1,0,0,0,0,0,0,0,4,0,0,0,0,0]
- Lookup first nibble in table, lookup second, compute bitwise AND.

Array of nibbles:

- original: [a0 a1 a2 a3 a4 ...]
- shift: [a1 a2 a3 a4 ...]
- shift: [a2 a3 a4 ...]
- f([a0 a1 a2 a3 a4 ...]) AND g([a1 a2 a3 a4 ...]) AND g([a2 a3 a4 ...])

UTF-16 to UTF-8

The other direction (from UTF-16 to UTF-8) is somewhat easier!

UTF-16 to UTF-8 (ASCII)

If all 16-bit words are ASCII (0000-007F), use a fast routine: 16 bytes into 8 'packed' bytes.

UTF-16 to UTF-8 (0000-07FF)

If all 16-bit words are in (0000-07FF)... build an 8-bit bitset indicating which 16-byte words are ASCII (0000-007F), load a shuffle mask, permute and patch.

UTF-16 to UTF-8 (0000-07FF, E000-FFFF)

If all 16-bit words are in the ranges 0000-D7FF, E000-FFFF, we use another similar specialized routine to produce sequences of one-byte, two-byte and three-byte UTF-8 characters.

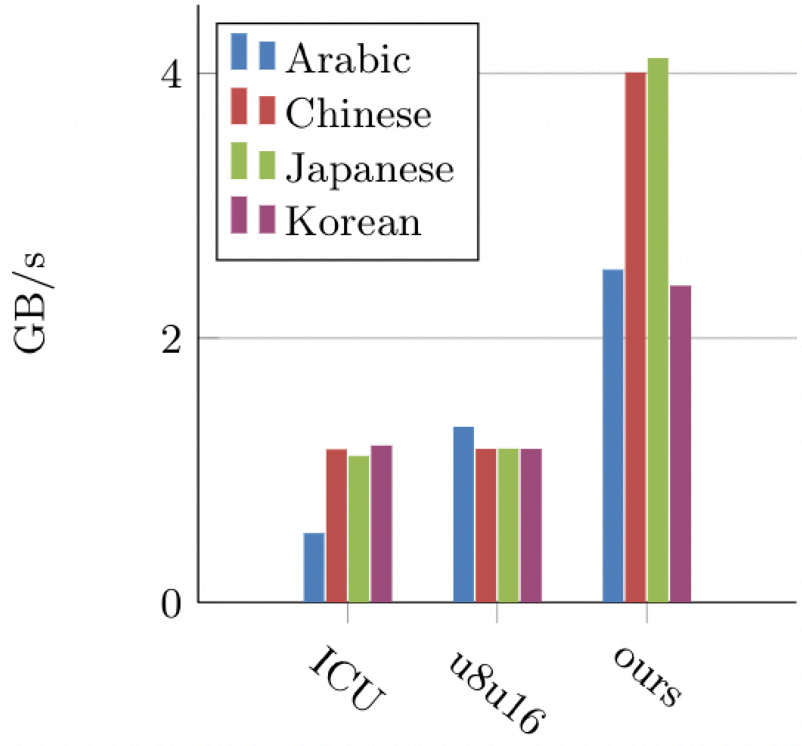
Otherwise, when we detect that the input register contains at least one part of a surrogate pair, we fall back to a conventional/scalar code path.

Experiments

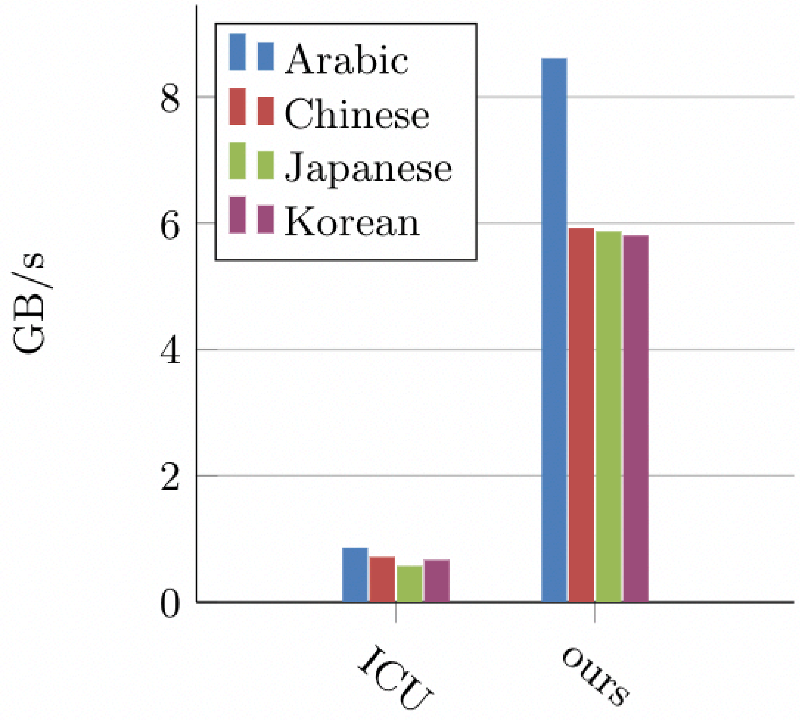
- AMD processor (AMD EPYC 7262, Zen 2 microarchitecture, 3.39 GHz) and GCC10.
- International Components for Unicode (UCI)
- u8u16 library
- lipsum text in various languages

ASCII transcoding

	UTF-8 to UTF-16	UTF-16 to UTF-8
simdutf	20 GB/s	36 GB/s
UCI	1 GB/s	2 GB/s



(a) UTF-8 to UTF-16 transcoding



(b) UTF-16 to UTF-8 transcoding

Software

https://github.com/simdutf/simdutf

- Open source, no patent.
- ARM NEON, SSE, AVX...
- Support runtime dispatch: adapts to your CPU.
- Easy to use: drop `simdutf.cpp` and `simdutf.h` in your project.
- Compiles to tens of kilobytes.

Further reading

- Lemire, Daniel and Wojciech Muła , Transcoding Billions of Unicode Characters per Second with SIMD Instructions, Software: Practice and Experience (to appear) https://r-libre.teluq.ca/2400/
- Blog: https://lemire.me/blog/